Worksheet 2.2 — Product & Quotient Rules

Grade 12 Calculus and Vectors (MCV4U) · Unit 2: Derivative Rules

The Big Ideas

Product: $(fg)' = f'g + fg'$. Quotient: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ top derivative first.

- $(fg)' = f'g + fg'$ (not $f'g'$).
- $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.
- If expanding is easy, expand and use the power rule.

The Big Picture

Products

The derivative of a product is *not* the product of derivatives: $\frac{d}{dx}[fg] = f'g + fg'$.

Quotients

For a quotient, $\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$ mind the order in the numerator.

Choosing a rule

Identify the structure first: a product calls for the product rule, a fraction for the quotient rule.

Worked Examples

Example 1 — Product rule

Differentiate $f(x) = x^2(x+1)$.

Solution. **Step 1 (identify the two factors):** $u = x^2$ (so $u' = 2x$) and $v = x + 1$ (so $v' = 1$).
Step 2 (apply the product rule $u'v + uv'$): $2x(x+1) + x^2(1)$.
Step 3 (expand and simplify): $2x^2 + 2x + x^2 = 3x^2 + 2x$.
Conclusion: $f'(x) = 3x^2 + 2x$ (check by expanding first: $x^2(x+1) = x^3 + x^2 \to 3x^2 + 2x$, matching).

Example 2 — Two factors

Differentiate $f(x) = (2x+1)(x^2 - 3)$.

Solution. **Step 1 (identify the two factors):** $u = 2x + 1$ (so $u' = 2$) and $v = x^2 - 3$ (so $v' = 2x$).
Step 2 (apply the product rule): $2(x^2 - 3) + (2x+1)(2x)$.
Step 3 (expand): $2x^2 - 6 + 4x^2 + 2x$.
Conclusion: $f'(x) = 6x^2 + 2x - 6$.

Example 3 — Quotient rule

Differentiate $f(x) = \dfrac{x+1}{x-1}$.

Solution. **Step 1 (identify top and bottom):** $u = x+1$ (so $u' = 1$) and $v = x-1$ (so $v' = 1$).

Step 2 (apply the quotient rule $\frac{u'v-uv'}{v^2}$): $\dfrac{(1)(x-1)-(x+1)(1)}{(x-1)^2}$.

Step 3 (simplify the numerator): $(x-1)-(x+1) = -2$.

Conclusion: $f'(x) = \dfrac{-2}{(x-1)^2}$.

Example 4 — Polynomial over linear

Differentiate $f(x) = \dfrac{x^2}{x+1}$.

Solution. **Step 1 (identify top and bottom):** $u = x^2$ (so $u' = 2x$) and $v = x+1$ (so $v' = 1$).

Step 2 (apply the quotient rule): $\dfrac{2x(x+1)-x^2(1)}{(x+1)^2}$.

Step 3 (simplify the numerator): $2x^2+2x-x^2 = x^2+2x$.

Conclusion: $f'(x) = \dfrac{x^2+2x}{(x+1)^2}$.

Example 5 — From a table

Given $f(2) = 3, f'(2) = 1, g(2) = 4, g'(2) = -2$, find $(fg)'(2)$.

Solution. **Step 1 (recall the product rule):** $(fg)' = f'g + fg'$.

Step 2 (substitute the given values at $x = 2$): $(1)(4)+(3)(-2)$.

Conclusion: $= 4-6 = -2$.

Example 6 — Product rule

Differentiate $f(x) = x^3(x-2)$.

Solution. **Step 1 (identify the two factors):** $u = x^3$ (so $u' = 3x^2$) and $v = x-2$ (so $v' = 1$).

Step 2 (apply the product rule): $3x^2(x-2)+x^3(1)$.

Step 3 (expand and combine): $3x^3-6x^2+x^3$.

Conclusion: $f'(x) = 4x^3-6x^2$.

Example 7 — Two factors

Differentiate $f(x) = (x+2)(x^2+1)$.

Solution. **Step 1 (identify the two factors):** $u = x+2$ (so $u' = 1$) and $v = x^2+1$ (so $v' = 2x$).

Step 2 (apply the product rule): $(1)(x^2+1)+(x+2)(2x)$.

Step 3 (expand and combine): x^2+1+2x^2+4x.

Conclusion: $f'(x) = 3x^2+4x+1$.

Example 8 — Quotient rule

Differentiate $f(x) = \dfrac{x}{x+2}$.

Solution. **Step 1 (identify top and bottom):** $u = x$ (so $u' = 1$) and $v = x + 2$ (so $v' = 1$).

Step 2 (apply the quotient rule): $\dfrac{(1)(x+2) - x(1)}{(x+2)^2}$.

Step 3 (simplify the numerator): $(x + 2) - x = 2$.

Conclusion: $f'(x) = \dfrac{2}{(x+2)^2}$.

Example 9 — Quotient rule

Differentiate $f(x) = \dfrac{x-1}{x+1}$.

Solution. **Step 1 (identify top and bottom):** $u = x - 1$ (so $u' = 1$) and $v = x + 1$ (so $v' = 1$).

Step 2 (apply the quotient rule): $\dfrac{(1)(x+1) - (x-1)(1)}{(x+1)^2}$.

Step 3 (simplify the numerator): $(x + 1) - (x - 1) = 2$.

Conclusion: $f'(x) = \dfrac{2}{(x+1)^2}$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = x^2(x-3)$.

Question 2

Differentiate $f(x) = (3x-1)(x+2)$.

Question 3

Differentiate $f(x) = \dfrac{x+2}{x-2}$.

Question 4

Differentiate $f(x) = \dfrac{x^2}{x-1}$.

Question 5

With $f(1) = 2, f'(1) = 3, g(1) = 5, g'(1) = -1$, find $(fg)'(1)$.

Question 6

Differentiate $f(x) = x^4(2x + 1)$.

Question 7

Differentiate $f(x) = (x^2 + 1)(x - 4)$.

Question 8

Differentiate $f(x) = \dfrac{2x}{x + 3}$.

Question 9

Differentiate $f(x) = \dfrac{1}{x^2 + 1}$ (quotient rule).

Question 10

Differentiate $f(x) = x(x^2 - 5)$.

Question 11

With $f(0) = 1, f'(0) = 2, g(0) = 3, g'(0) = 4$, find $\left(\frac{f}{g}\right)'(0)$.

Question 12

Differentiate $f(x) = (x - 1)(x + 1)$ two ways (rule and expansion).

Question 13 — Challenge

Find the tangent to $f(x) = \dfrac{x}{x+1}$ at $x = 1$.

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = x^2(x-3)$.
2. Differentiate $f(x) = (3x-1)(x+2)$.
3. Differentiate $f(x) = \dfrac{x+2}{x-2}$.
4. Differentiate $f(x) = \dfrac{x^2}{x-1}$.
5. With $f(1)=2, f'(1)=3, g(1)=5, g'(1)=-1$, find $(fg)'(1)$.
6. Differentiate $f(x) = x^4(2x+1)$.
7. Differentiate $f(x) = (x^2+1)(x-4)$.
8. Differentiate $f(x) = \dfrac{2x}{x+3}$.
9. Differentiate $f(x) = \dfrac{1}{x^2+1}$ (quotient rule).
10. Differentiate $f(x) = x(x^2-5)$.
11. With $f(0)=1, f'(0)=2, g(0)=3, g'(0)=4$, find $\left(\frac{f}{g}\right)'(0)$.
12. Differentiate $f(x) = (x-1)(x+1)$ two ways (rule and expansion).
13. Find the tangent to $f(x) = \dfrac{x}{x+1}$ at $x=1$.

✓ Answer Key

1. $3x^2 - 6x$
2. $6x + 5$
3. $\dfrac{-4}{(x-2)^2}$
4. $\dfrac{x^2-2x}{(x-1)^2}$
5. $(3)(5)+(2)(-1) = 13$
6. $10x^4 + 4x^3$
7. $3x^2 - 8x + 1$
8. $\dfrac{6}{(x+3)^2}$
9. $\dfrac{-2x}{(x^2+1)^2}$
10. $3x^2 - 5$
11. $\dfrac{(2)(3)-(1)(4)}{9} = \frac{2}{9}$
12. both give $2x$
13. $f'(1) = \frac{1}{4}$, point $(1, \frac{1}{2})$: $y = \frac{1}{4}x + \frac{1}{4}$